MANURANGA JAYAWARDHANA

SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Computer Engineering Graduate (B.Sc. Eng. Hons) and Software Engineer specializing in Full Stack development with Java and Spring Boot. Proficient in designing scalable solutions using SQL, RESTful APIs, and modern JavaScript frameworks like React. Skilled in transforming complex requirements into efficient code, with a strong foundation in database management and software engineering principles.

EDUCATION

FACULTY OF ENGINEERING, UNIVERSITY OF JAFFNA (2021 – 2025)
B. Sc. Eng. (Hons) in Computer Engineering

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming (OOP), Software Engineering, Database Management Systems, Operating Systems, Computer Networks, Distributed Systems, Artificial Intelligence, Embedded Systems, Computer Architecture, Machine Learning, Deep Learning.

SKILLS SUMMARY

- **Programming Languages**: Java, JavaScript, TypeScript, Python
- **Frameworks/Libraries:** Spring Boot, Hibernate (JPA), Spring MVC, Node.js, Serverless, NestJS, Next.js, React.js
- **Database & Caching**: SQL (MySQL, PostgreSQL), NoSQL (MongoDB), Redis
- **Web Development**: HTML, CSS, Tailwind CSS
- **Architecture & Messaging:** Apache Kafka, RabbitMQ, Serverless, RESTful APIs
- **Cloud & DevOps**: AWS (Amazon Web Services), Google Cloud, Docker, Git, GitHub
- **Testing**: Postman, Junit, Jest, k6

WORK EXPERIENCE

Intern Software Engineer
Sayanu (Pvt) Ltd (Feb 2025 – Aug 2025)

- Engineered scalable e-commerce platforms using **NestJS** and the **Serverless Framework** for the backend, paired with **Next.js** and **React** for responsive frontend interfaces.
- Orchestrated cloud deployment and monitoring on **AWS**, utilizing **Lambda** for serverless execution, **Amplify** for frontend hosting, **S3** for static resource management, and **CloudWatch** for real-time logging.
- Implemented secure authentication and authorization(Role based access control) protocols using **JWT** (JSON Web Tokens) to ensure robust data protection and user access control.
- Managed relational database operations using **PostgreSQL**, optimizing schemas and queries within a collaborative **Agile** development environment.

PROJECTS

Bidflare - Online Bidding Platform | GitHub

- Developed a secure backend using **Java**, **Spring Boot**, **Spring Security**, and **JWT** for authentication, while optimizing data persistence and integrity using **Hibernate (JPA)** and **PostgreSQL**.

- Engineered a dynamic frontend interface with **Next.js** and **Tailwind CSS**, utilizing **Redux** and **Context API** to manage complex application state efficiently.
- Implemented **CI/CD** pipelines to automate the deployment process, containerizing services with **Docker** and hosting the production build on an **AWS EC2** instance.

FormPilot - AI-Powered Fitness Analysis Platform| GitHub

- Designed a scalable **Microservices architecture** using **Java** and **Spring Boot**, implementing **Eureka Server** for service discovery and **API Gateway** for centralized request routing.
- Integrated **Gemini API** within the backend to generate AI-driven performance insights, utilizing **RabbitMQ** and **Kafka** with **Spring Cloud Stream** for asynchronous messaging and **WebClient** for synchronous inter-service communication.
- Secured the ecosystem using **Keycloak** for identity management, enforcing **OAuth2** with **PKCE** and **JWT** standards to ensure robust authentication and authorization.
- Developed a dynamic **React** frontend using **Redux** for state management and **Material UI**, while managing data across a hybrid **PostgreSQL** and **MongoDB** database structure.

BlitzScale - High-Concurrency Flash Sale Engine| GitHub

- Architected an event-driven microservices platform using **Java** and **Spring Boot**, integrating **Eureka Server** for service discovery, **Spring Cloud Gateway** for centralized routing, and **Keycloak (OAuth2/JWT)** for secure identity management.
- Engineered a resilient checkout mechanism utilizing **Redis** for distributed rate-limiting and sub-millisecond atomic stock locking, paired with **Apache Kafka** for asynchronous order processing to prevent database deadlocks.
- Benchmarked local system scalability using **k6** under local hardware constraints, successfully processing **39,300+ orders in 50 seconds (800 concurrent users)** to achieve a sustained **throughput of 760+ TPS** with a **511ms average latency and 0.00% failure rate** across **PostgreSQL** and **H2** databases.

CERTIFICATIONS

- Introduction to Web Development with HTML, CSS, JavaScript — 2023 IBM - Coursera
- Developing Front-End Apps with React — 2024 IBM – Coursera
- Building Scalable Java Microservices with Spring Boot and Spring Cloud — 2024 Google Cloud - Coursera

EXTRA CURRICULAR ACTIVITIES

- Computer Engineering Society member at University of Jaffna
- Engineering Student Union Executive Member
- Cricket team member at University of Jaffna

SOFT SKILLS

- Adaptability, Problem Solving, Team Collaboration, Critical Thinking, Time Management, Leadership

REFEREES

- **Mr. Waruna Sri Dhanapala** (SLAS - Special Grade) Secretary (Actg.), Ministry of Digital Economy Chairman, Telecommunications Regulatory Commission of Sri Lanka (TRCSL)
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- **Mr. J.M. Sarath Ambegoda** Director of Engineering at Proleadsoft USA, Co-Founder & Software Architect at Sayanu (Pvt) Ltd
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